

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TRANSFORMER MECHANICS WORKSHEET

Course Code:	CSA6501	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i>		
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Section / Batch:	CSE-AI	Date:	

Instructions to Students

- Answer one questions. It carries 100 Marks. Total: 100 Marks.
- Tokenize a sentence.
- Convert tokens to token IDs.
- Explain embeddings.
- Label a Transformer encoder.
- Identify Query, Key, and Value.
- Explain which words receive higher attention.
- Compare Transformer models.
- Complete a Transformer pipeline.
- Draw a block diagram.

Code of Conduct

I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **A. Sai Rohit**

SECTION A – TRANSFORMER MECHANICS WORKSHEET

Question: Sequence Representation Worksheet

1. Tokenizing a Sentence

Sentence:

"The cat sat on the mat."

Tokens:

["The", "cat", "sat", "on", "the", "mat", "."]

2. Converting Tokens to Token IDs

Assume the vocabulary is:	
Token	Token ID
The	101
cat	205
sat	330
on	115
the	102
mat	450
.	103

Token IDs:

[101, 205, 330, 115, 102, 450, 103]

These IDs are numerical representations that the Transformer processes.

3. Explaining Embeddings

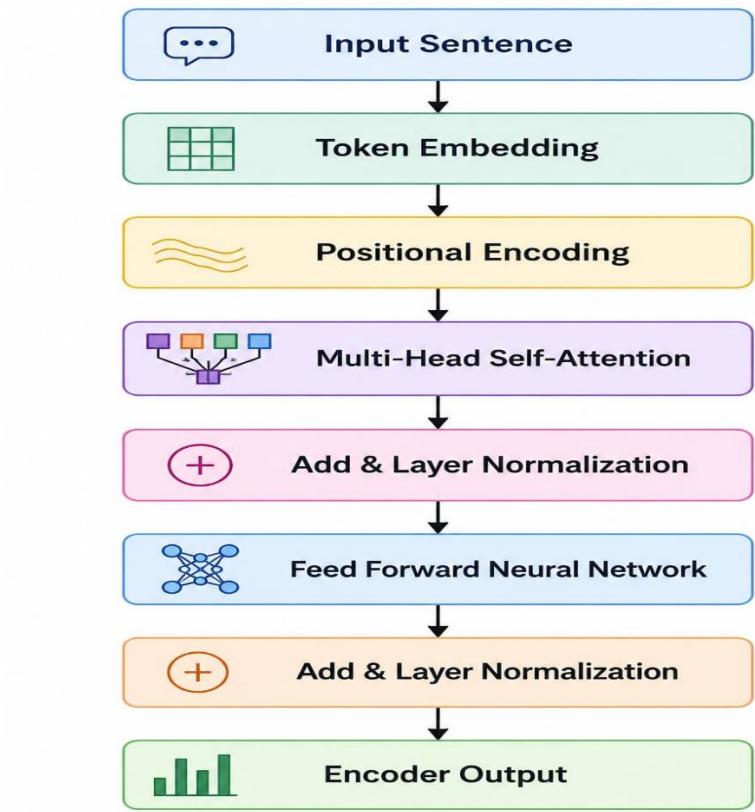
Embeddings are numerical vector representations of words. They convert each token ID into a list of decimal numbers so that the Transformer can understand the meaning and relationship between words.

Token	Embedding Vector (Example)
The	[0.21, 0.45, -0.32, 0.81]
cat	[0.72, -0.16, 0.59, 0.34]
sat	[0.18, 0.63, -0.41, 0.52]
on	[0.40, -0.22, 0.33, 0.75]
the	[0.25, 0.48, -0.30, 0.80]
mat	[0.61, 0.12, -0.55, 0.43]

Advantages of Embeddings

- Convert words into numerical vectors.
- Capture the meaning of words.
- Similar words have similar vectors.
- Help the Transformer understand context.

4. Labelling a Transformer Encoder



Explanation of Each Block

Component	Function
Input Sentence	Original text given to the model.
Token Embedding	Converts tokens into vectors.
Positional Encoding	Adds word position information.
Multi-Head Self-Attention	Finds relationships between words.
Add & Layer Normalization	Stabilizes training and improves learning.
Feed Forward Network	Processes features further.
Encoder Output	Final contextual representation of words.

5. Identifying Query (Q), Key (K), and Value (V)

Sentence:

The cat sat on the mat.

When processing the word "**sat**", the Transformer creates three vectors.

Query (Q)

- Represents what the current word is looking for.
- Generated from the current word.

Example:

Word = sat

Q = "Which words are related to me?"

Key (K)

- Represents the information each word contains.
- Every word has its own Key.

Example:

The → K

cat → K

sat → K

on → K

the → K

mat → K

Value (V)

- Stores the actual information of each word.
- Used to produce the final output.

Example:

The → V

cat → V

sat → V

on → V

the → V

mat → V

Attention Formula

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Simple Explanation

- **Query** asks: *Which words are important to me?*
- **Key** answers: *Here is the information I have.*
- **Value** provides: *This is the information passed to the output.*

6. Explanation of Which Words Receive Higher Attention

Sentence:

The cat sat on the mat.

When processing the word "**sat**", attention is given as follows:

Word	Attention Level	Reason
The	Medium	Refers to the subject.
cat	High	Subject performing the action.
sat	Highest	Current word being processed.
on	Medium	Connects the action to the object.
the	Low	Less important.
mat	Medium	Object related to the action.

Conclusion

The word "**cat**" receives high attention because it is the subject of "**sat**". The word "**mat**" also receives attention because it is related to where the cat sat.

7. Comparison of Transformer Models

Model	Encoder	Decoder	Main Application
BERT	Yes	No	Text Classification, Question Answering
GPT	No	Yes	Text Generation
T5	Yes	Yes	Translation, Summarization
RoBERTa	Yes	No	Improved Text Understanding
BART	Yes	Yes	Summarization and Translation

Comparison

BERT

- Uses only the encoder.
- Reads text in both directions.
- Best for understanding text.

GPT

- Uses only the decoder.
- Generates text one word at a time.
- Best for text generation.

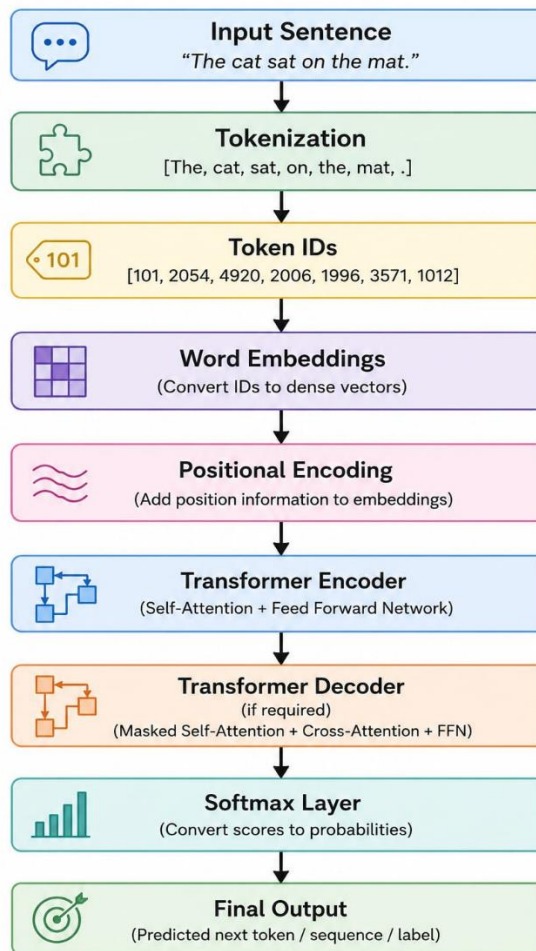
T5

- Uses both encoder and decoder.
- Converts every NLP task into a text-to-text format.

8. Transformer Pipeline

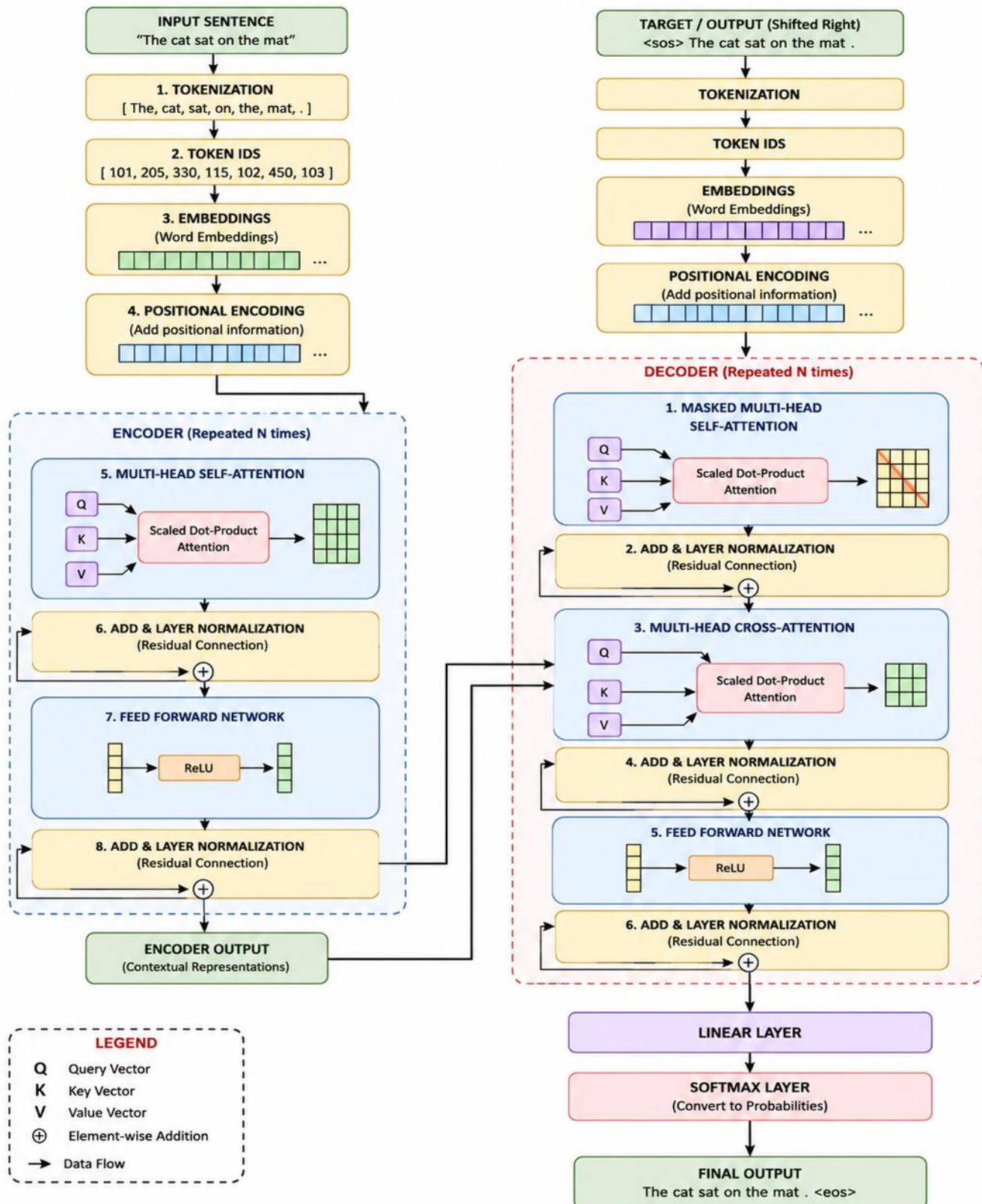
Pipeline Steps

1. Input sentence is given.
2. Sentence is tokenized.
3. Tokens are converted to IDs.
4. IDs become embeddings.
5. Positional encoding is added.
6. Encoder processes the sequence.
7. Decoder generates output (if needed).
8. Softmax predicts the final output.



9. Block Diagram

TRANSFORMER MODEL – COMPLETE BLOCK DIAGRAM



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____